Muhammad Huzaifa

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About

Software Engineer with a experience in fintech and defense systems. Currently pursuing a Master's in Computer Science & Engineering at the University of Oulu, specializing in Cyber Security. Experienced in architecting large scale systems,with expertise in full stack development, system integration, and DevOps.

Technical Skills

- **Languages:** Java, C#, Python, Go, JavaScript, TypeScript, SQL, C, C++.
- **Frameworks & Libraries:** Spring Boot, ASP.NET/Core, React JS, Node JS, Hibernate
- **Databases:** PostgreSQL, MSSQL, MySQL, YugaByteDB.
- **Cloud & DevOps:** Azure (Virtual Network, Azure Kubernetes Service, KeyVault, Storage Browser, Virtual Machines), Google Cloud, Docker, Kubernetes, Jenkins, GitHub Actions.
- **Tools & Platforms:** IntelliJ, Git, Visual Studio, Visual Studio Code, Google Colab, Postman, Jira.
- **Software Engineering:** Object Oriented Programming, Agile, Scrum, Technical Documentation.
- **Security & Compliance:** PCI-DSS, GDPR, OWSAP T10.

Education

University of Oulu — **September 2025 – June 2027**
Master of Science in Computer Science & Engineering — *Oulu, Finland*

- **Specialization Track:** Cyber Security; Software and Data Security
- **Relevant Courses:** Professional Software Engineering Processes & Human Factors, Distributed Systems,Advanced Software Quality & Security, Software Project, International Crisis Management, Data Mining, Cryptographic Systems & their Weaknesses, Security Engineering.

National University of Sciences & Technology — **August 2020– May 2024**
Bachelors of Science in Computer Science — *Islamabad, Pakistan*

- **Relevant Courses:** Object Oriented Programming, Data Structures & Algorithms, Database Systems, Web Engineering, Software Engineering, Information Security, Distributed Computing, Artificial Intelligence, Deep Learning, Operating Systems, Computer Networks.
- **Thesis:** OptiGuard: Generalized, Attention-Driven & Explainable Glaucoma Classification (IEEE)

Experience

Research Intern — **June 2026– Present**
University of Oulu — *Oulu, Finland*

- Extending the open-source AVISE (AI Vulnerability Identification & Security Evaluation) framework to support safety and robustness assessments of Text-to-Image (Txt2Image) generative models.
- Designed and executed custom Security Evaluation Tests (SETs) using Python to evaluate model resistance against adversarial, harmful, and evasive prompt injection attacks.
- Co-authoring a forthcoming research paper to formalize methodology, benchmark findings, and security vulnerabilities discovered during Txt2Image evaluations.
- **Technologies:** Python.
- **Tools:** Docker, Git, Google Colab.

Full Stack Developer — **February 2024– March 2026**
Datapulse Technologies (FinTech) — *Ireland (Remote)*

- Maintained and optimized a legacy payment architecture used as a whitelabel solution across multiple regions; improved performance by 25% and reduced latency to ensure stability under peak production loads.
- Architected and developed a high efficiency replacement system, addressing functional gaps in the legacy codebase.
- Integrating the payment solution with major global acquirers such as E-Comprocessing, Planet and Rapyd, and digital wallets like Apple Pay and Google Pay, enabling seamless and secure payment processing across regions.
- Successfully integrated independent physical terminal systems into a unified gateway for card-present and card-not-present transactions.

- Maintained a scalable microservices architecture to process thousands of daily transactions with high reliability and fault tolerance.
- Managing CI/CD pipelines and cloud deployments using Docker, Kubernetes, and Jenkins to ensure high availability, (GitHub → Jenkins → Docker → Kubernetes) enabling repeatable, secure, zero downtime releases just by a git push.
- Collaborated with compliance teams on PCI-DSS readiness; implemented secure data handling, access controls, and audit logging measures using SIEM and Cloud Defender.
- **Technologies:** Java (Spring Boot), Hibernate, C#, ASP.NET Core, React JS, Python.
- **Tools:** Azure, Docker, Kubernetes, Jenkins, PostgreSQL, MSSQL, YugabyteDB, Git.

Trainee Engineer **July 2022– May 2024**
Rapidev (DefTech) *Islamabad, Pakistan*

- Contributed to RapiGuard, an Anti-UAV defense system for real time detection and neutralization of aerial threats.
- Integrated hardware sensors (detectors, jammers, spoofers) with software control applications for real time response.
- Developed a Key Management Application for secure radio communications, managing the lifecycle of encryption keys for mission critical hardware.
- Engineered the injection of cryptographic keys into radio units via the UART protocol, ensuring secure and reliable data transfer between software and hardware.
- Developed a Tactical Situational Awareness mobile application using Flutter, delivering real time geospatial data visualization for Android.
- **Technologies:** C#, Asp.Net/ Core, Python, React JS, Flutter, SQL, UART.
- **Tools:** Visual Studio, Visual Studio Code, Git, GitHub.

Software Engineering Intern **March 2023– August 2023**
Bitnine *San Francisco, USA (Remote)*

- Contributed to Apache's open-source graph-based software solutions, enhancing functionality and performance.
- Developed the AG Cloud website to provide a user-friendly interface for accessing cloud-based graph solutions.
- Extended PostgreSQL's capabilities by contributing to AGE, a graph extension for advanced data handling.
- Technologies: C++, Go, React JS, Node JS, MongoDB, PostgreSQL.
- Tools: Visual Studio Code, Git, GitHub.

Project

Nafalytics **Founder**
Stock Portfolio Tracker

- **Nafalytics** — derived from "Nafa" (profit) and "analytics" — is a fully functional stock market portfolio tracking application built to fill the gap in the Pakistani market for reliable portfolio management tools.
- Built from scratch over 3 months, used by 250 users to track portfolios, real time profit/loss, and trade activity.
- Developed a comprehensive feature suite including real time P/L, historical performance analytics with visualizations, automated payouts, and a notification engine.
- Managed the full product lifecycle and cloud deployment using Java Spring Boot, React JS, and PostgreSQL, with automated CI/CD workflows via GitHub Actions across Azure.
- **Technologies:** Java (Spring boot), React JS, PostgreSQL.
- **Tools:** Azure, Google Cloud, GitHub, Git Actions, Visual Studio Code, IntelliJ.
- Deployed here: https://nafalytics.com

PBT Payment Gateway **Work Project**
Payment processing system

- Maintained and optimised a legacy payment architecture used as a white-label solution across multiple regions; improved performance by 25% and reduced latency to ensure stability under peak production loads.
- Architected and developed a high efficiency replacement system, addressing functional gaps in the legacy codebase
- Integrating the payment solution with major global acquirers such as E-Comprocessing, Planet and Rapyd, and digital wallets like Apple Pay and Google Pay, enabling seamless and secure payment processing across regions.
- Successfully integrated independent physical terminal systems into a unified gateway for card-present and card-not-present transactions.
- Maintained a scalable microservices architecture to process thousands of daily transactions with high reliability and fault tolerance.
- Managing CI/CD pipelines and cloud deployments using Docker, Kubernetes, and Jenkins to ensure high availability, (GitHub → Jenkins → Docker → Kubernetes) enabling repeatable, secure, zero downtime releases just by a git push.

- Collaborated with compliance teams on PCI-DSS readiness; implemented secure data handling, access controls, and audit logging measures using SIEM and Cloud Defender.
- **Links:** https://pbtgateway.com/, https://merchants.pbtpay.eu/WebSite/DeveloperDocs.aspx

RapiGuard **Work Project**

Anti UAV Defense System

- Worked on RapiGuard, an Anti-UAV system that integrates hardware devices like jammers, spoofers, and detectors to detect and neutralize drone threats.
- Integrated latest hardware devices to detect and neutralize drone threats in a field environment.
- Developed APIs for communication with third party softwares.
- Contributed to updates and improvements, enhancing the system's efficiency in detecting and neutralizing UAV threats.
- **Links:** https://rapidev.ae/rapiguard.html

Publications

OptiGuard: Generalized, Attention-Driven & Explainable Glaucoma Classification **IEEE**

Glaucoma Detection, Computer-Aided Diagnosis, Artificial Intelligence, Deep Learning *Bachelors Thesis*

- A generalized, attention-driven, and explainable glaucoma classification tool, aimed at assisting early diagnosis.
- Acheived state of the art results on G-1020 & SMDG datasets.
- Paper highlights the methodology, results, and implications of our work.
- Awarded 1st Best Project distinction, adjudged by industry experts for its potential impact and technical excellence.
- Presented for publication at the 47th Annual International Conference of the IEEE, Engineering in Medicine and Biology Society, which was held in Copenhagen, Denmark.
- IEEE: https://ieeexplore.ieee.org/document/11253669
- Get to know more here: https://linktr.ee/optiguard

Certifications

Junior Penetration Tester **December 2025**

TryHackMe

- Hands on experience with OWASP Top 10 vulnerabilities including SQL Injection (SQLi), Cross-Site Scripting (XSS), IDOR, and Command Injection.
- Skilled in Authentication Bypass, SSRF, and File Inclusion (LFI/RFI) exploitation and remediation.
- Proficient in Content Discovery, Subdomain Enumeration, and manual application review via Browser Developer Tools.
- Understanding of Race Conditions and directory traversal vulnerabilities.

Languages

- English – Fluent (C1/C2)
- Finnish – Beginner (A1)